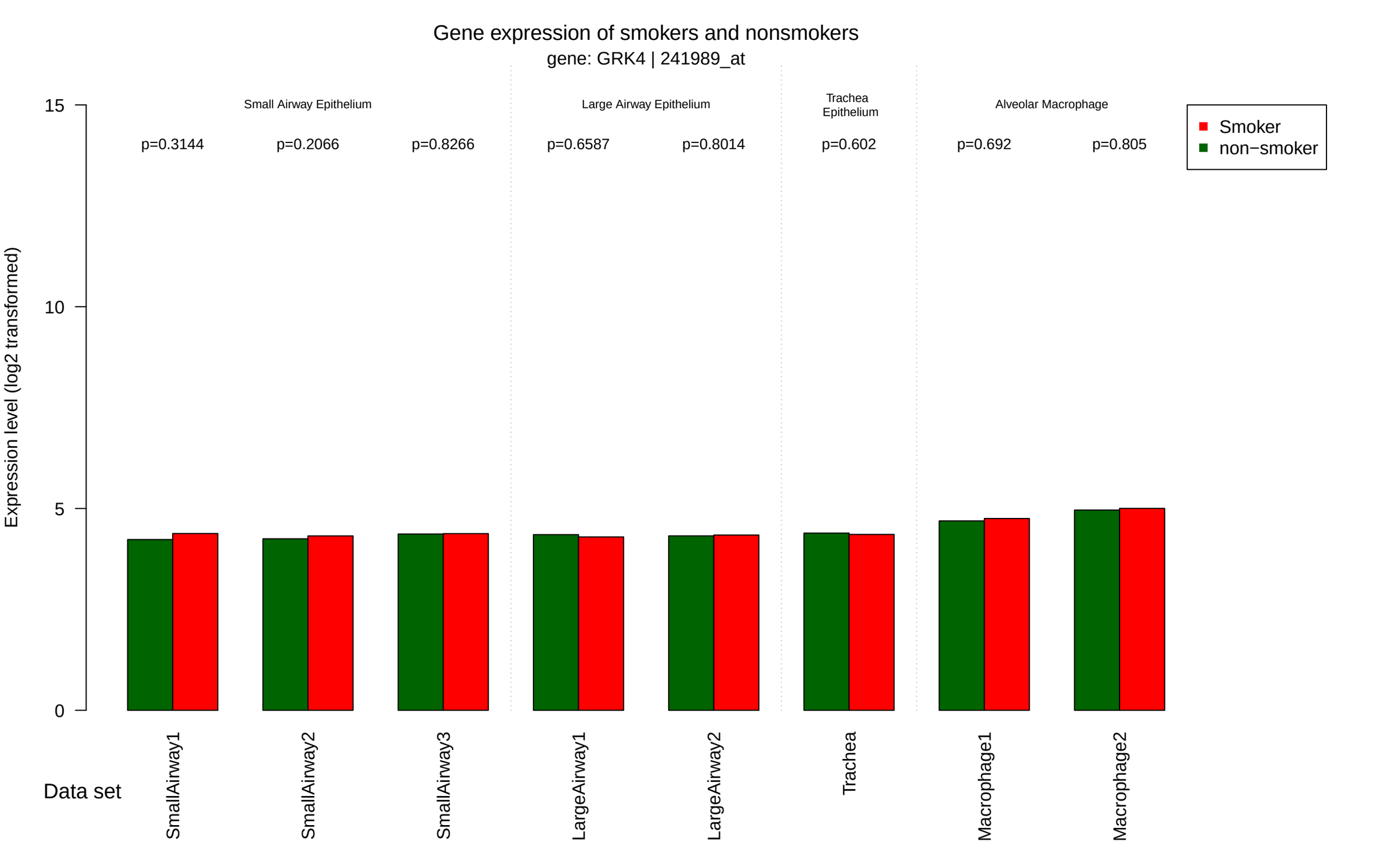
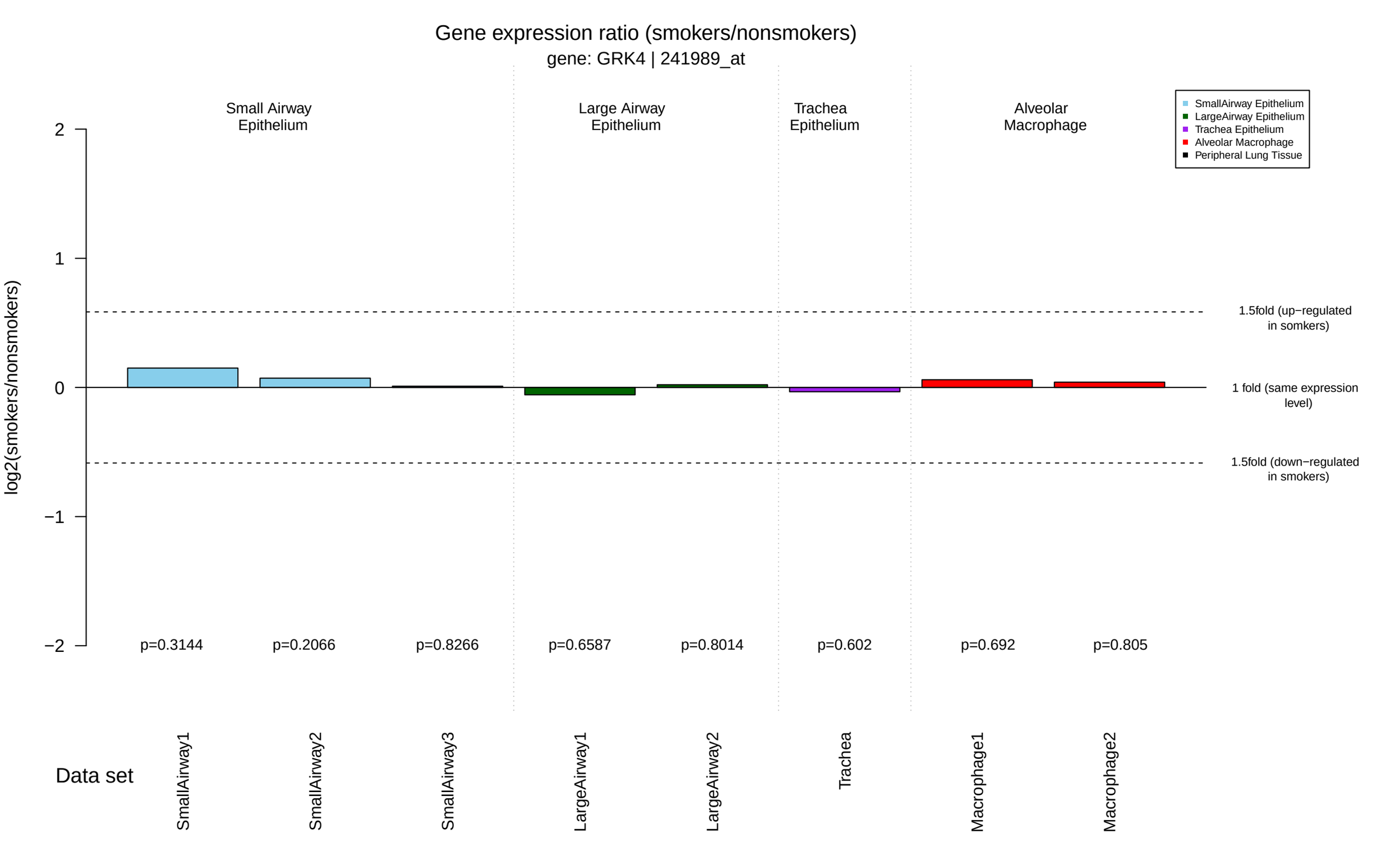
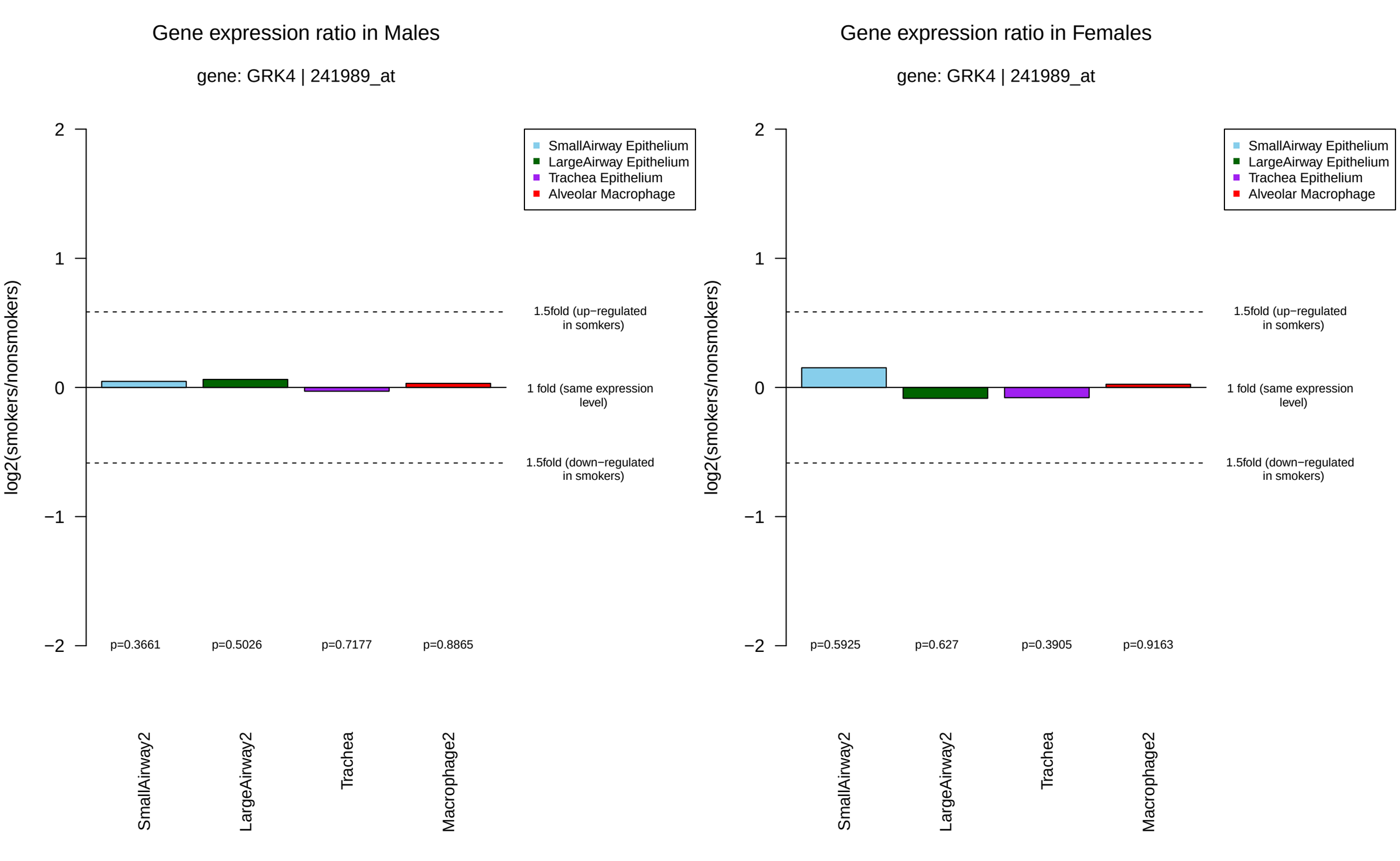
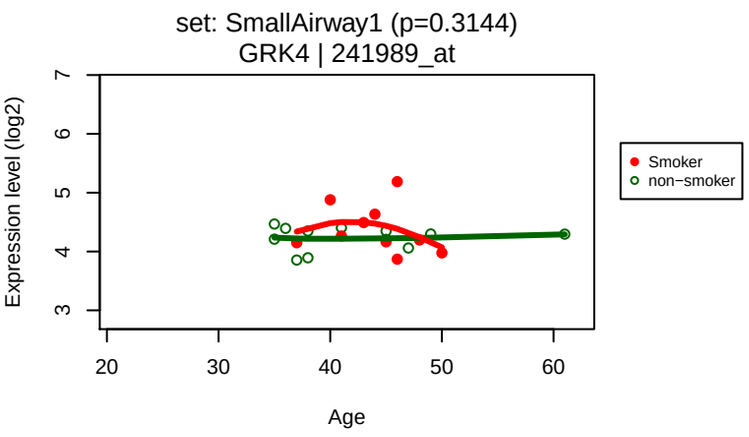
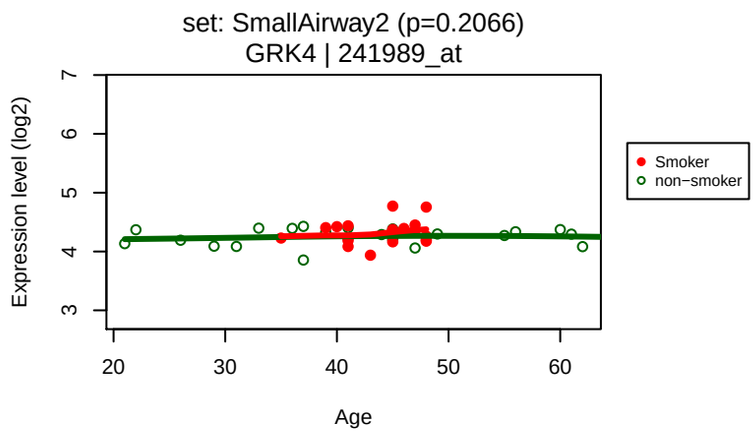
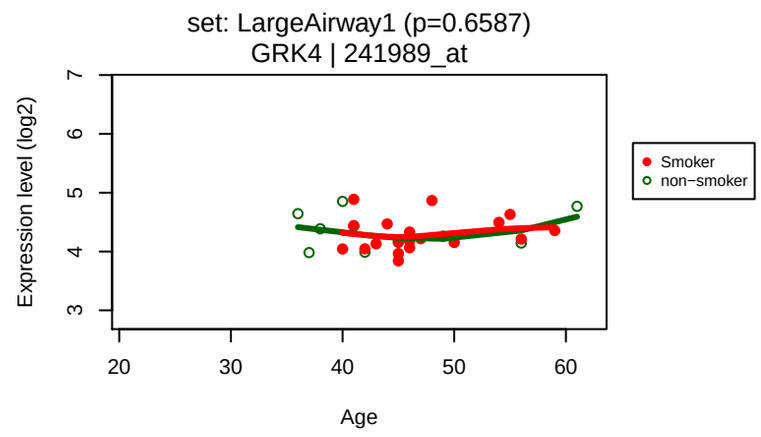

Gene expression of smokers and nonsmokers
gene: GRK4 | 241989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3144
p=0.2066
p=0.8266
p=0.6587
p=0.8014
p=0.602
p=0.692
p=0.805
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

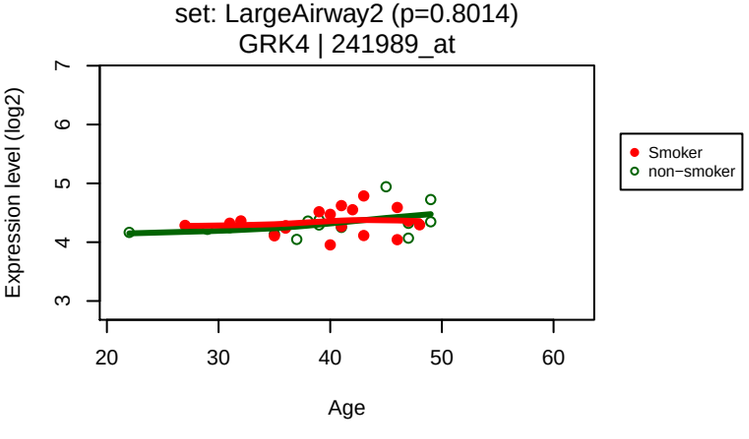

Gene expression ratio (smokers/nonsmokers)
gene: GRK4 | 241989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3144
p=0.2066
p=0.8266
p=0.6587
p=0.8014
p=0.602
p=0.692
p=0.805
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

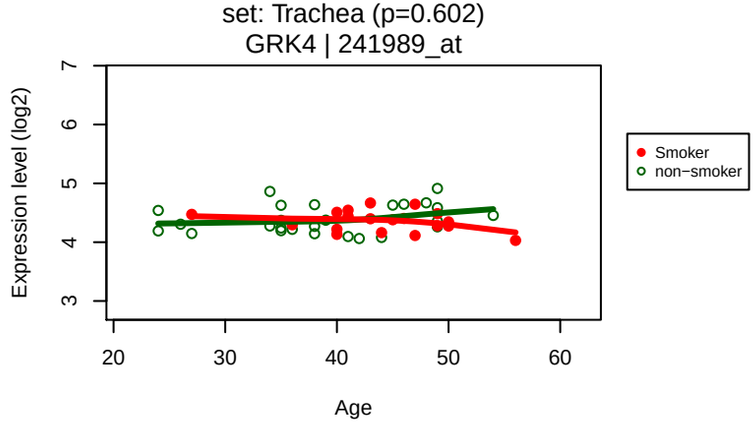

Gene expression ratio in Males
gene: GRK4 | 241989_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3661
p=0.5026
p=0.7177
p=0.8865
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRK4 | 241989_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5925
p=0.627
p=0.3905
p=0.9163
SmallAirway2
LargeAirway2
Trachea
Macrophage2

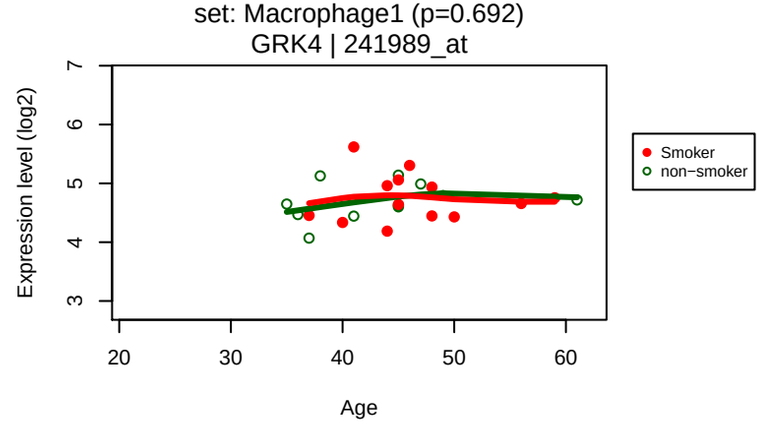

set: SmallAirway1 (p=0.3144)
GRK4 | 241989_at
Expression level (log2)
Age
Smoker
non−smoker

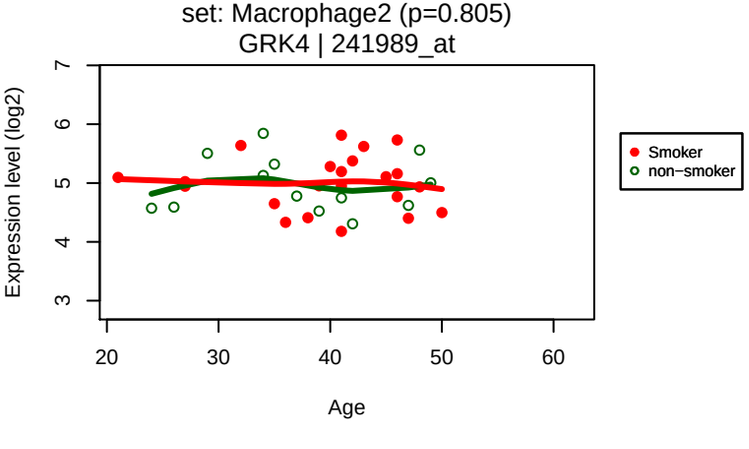

set: SmallAirway2 (p=0.2066)
GRK4 | 241989_at
Expression level (log2)
Age
Smoker
non−smoker

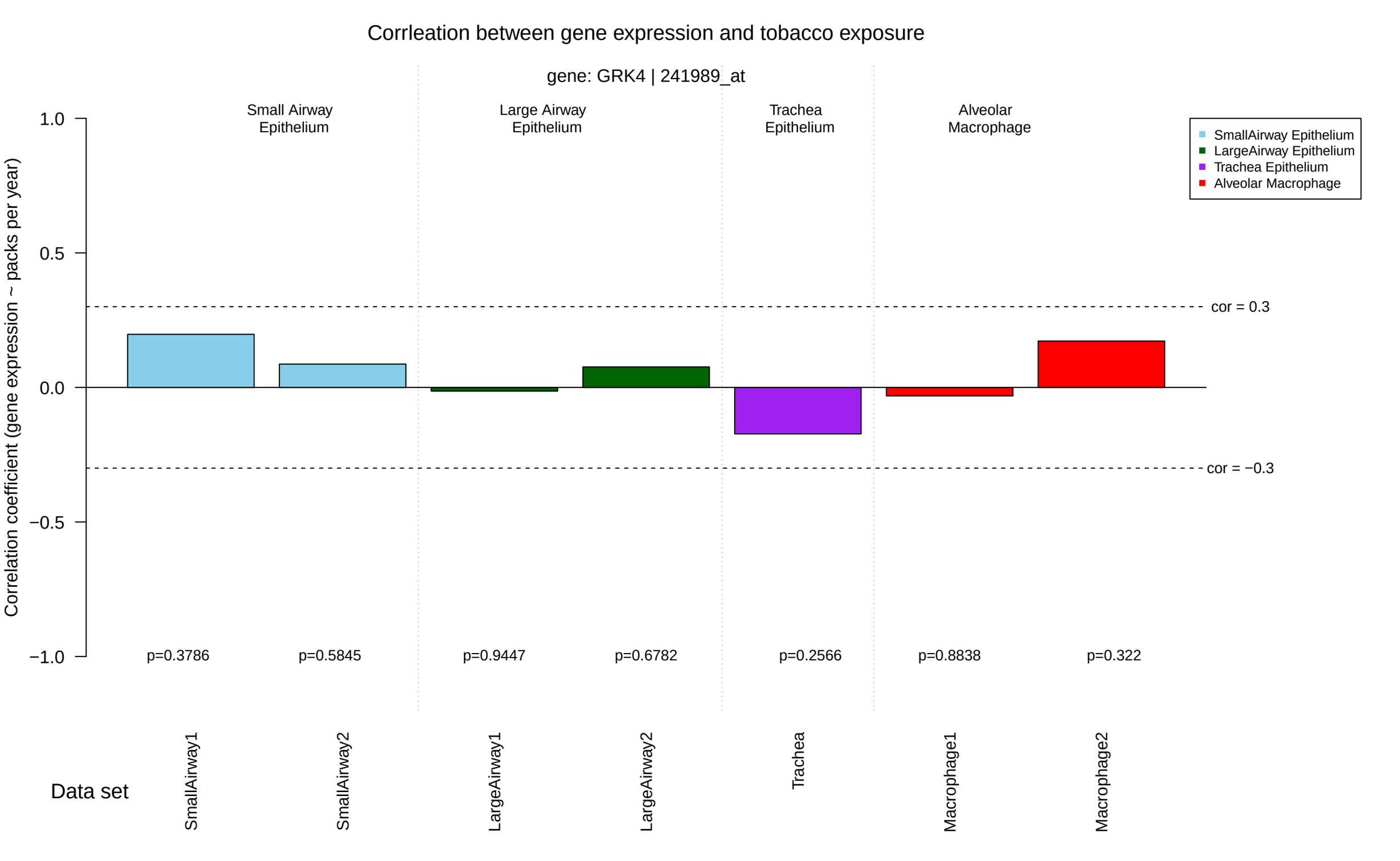

set: LargeAirway1 (p=0.6587)
GRK4 | 241989_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8014)
GRK4 | 241989_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.602)
GRK4 | 241989_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.692)
GRK4 | 241989_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.805)
GRK4 | 241989_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRK4 | 241989_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3786
p=0.5845
p=0.9447
p=0.6782
p=0.2566
p=0.8838
p=0.322
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2